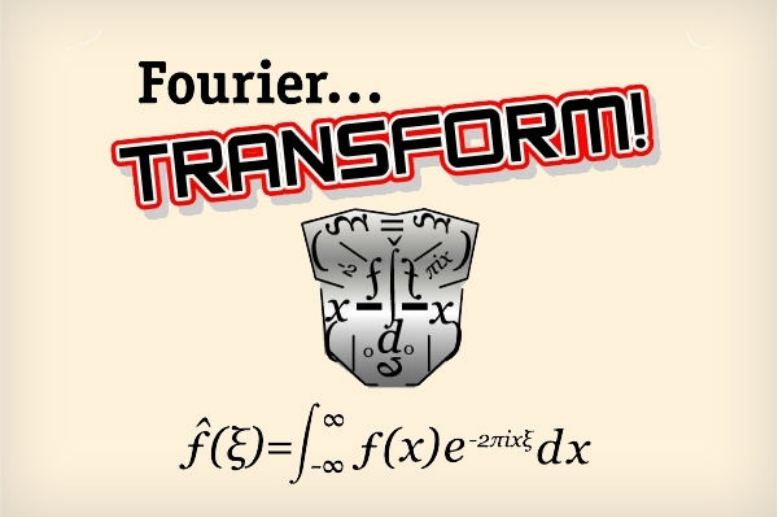
Signals and Systems

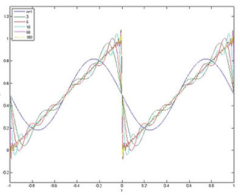
Module 6
The Fourier Transform



Overview

- This lecture will introduce the following topics:
  - Representation of aperiodic signals by the Fourier Integral
  - The Fourier Transform and the Inverse Fourier Transform
    - from the time-domain to the frequency-domain and back again
  - Properties of the Fourier Transform
  - Filtering and Windowing

Fourier Series Recap



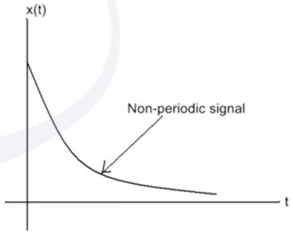
- (Almost) any periodic signal can be represented as an infinite sum of sinusoids via the Fourier series:

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \qquad D_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-jn\omega_0 t}dt$$

- Spectral plot
  - represent the magnitudes and phases of the harmonic terms of the Fourier series
  - to represent the frequency content of the signal
- Frequency-domain viewpoint
  - useful insights about a signal and how a system will effect it

Aperiodic Signal Representation

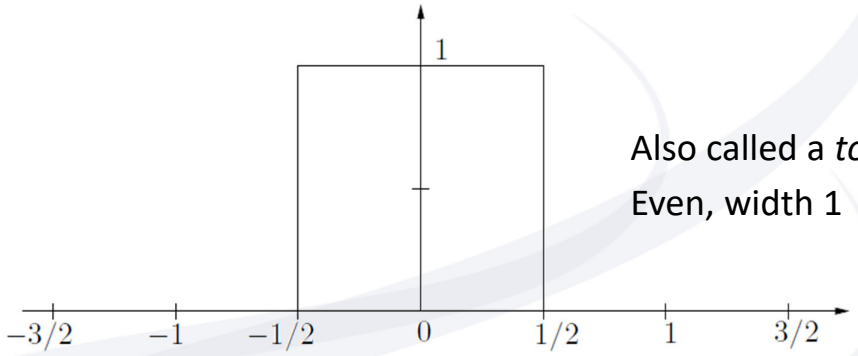
- Most real practical signals are not periodic
  - strict sense - periodic signal exists for all time - no real signals are periodic
- The Fourier series only applies to periodic signals
- ***Fourier transform*** allows us to transform any signal into the frequency-domain
- Shall show that an aperiodic signal can be expressed as a continuous sum (integral) of everlasting exponentials



Aperiodic Signal Representation

- Consider a straightforward rectangular pulse defined by:

$$x(t) = \text{rect}\,(t) = u\left(t + {}^{1}/_{2}\right) - u\left(t - {}^{1}/_{2}\right)$$



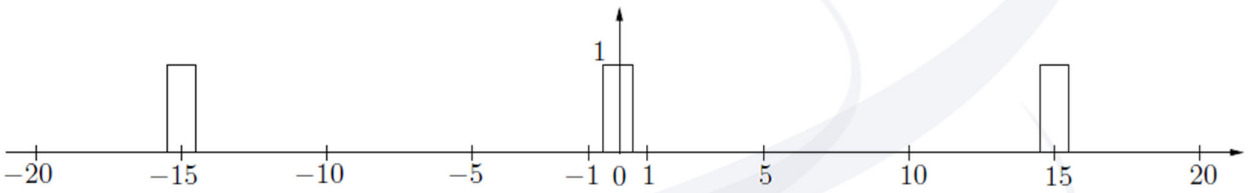
Also called a *top hat function*

Even, width 1

- Does this signal have a Fourier series?

Aperiodic Signal Representation

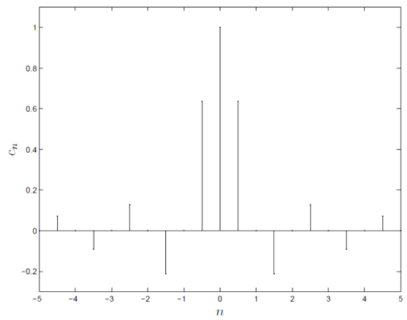
- Can we periodise it?
- Yes, repeat the non-zero part at intervals of T_0 along the time axis
  - where T_0 is arbitrarily large enough to ensure no overlaps



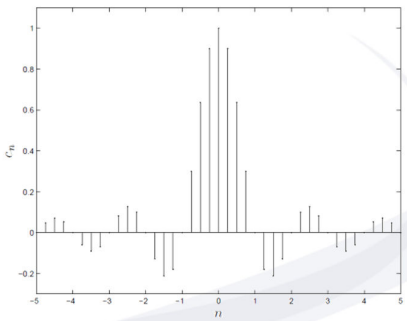
- Type of signal
  - e.g. from flipping a switch repeatedly on and off for a short period
  - e.g. pulse width modulation

Aperiodic Signal Representation

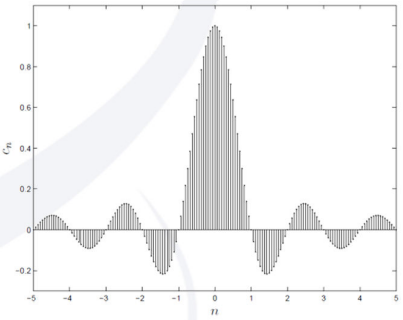
- The Fourier series now exists
  - Some normalised magnitude spectral plots of the Fourier coefficients:



Period = 2s



Period = 4s



Period = 16s

- What is happening as the period is increased?

Aperiodic Signal Representation

- Increasing T_0 means decreasing ω_0
  - so the harmonics become closer and closer
- Fourier coefficients for a rectangular pulse train given by:

$$D_n = \frac{1}{n\pi}\sin\left(\frac{n\pi}{T_0}\right)$$

- The points on the spectrum are $^{n}/_{T_0}$ apart
- What happens as $T_0 \to \infty$
  - i.e. representing our aperiodic signal as a periodic signal with a period of ∞
- What happens to D_n as $T_0 \to \infty$?

Aperiodic Signal Representation

- As $T_0 \to \infty$

$$\frac{1}{n\pi}\sin\left(\frac{n\pi}{T_0}\right) \to \frac{1}{T_0} \qquad (\sin\theta \approx \theta \text{ for small } \theta)$$

- Doubling T_0 halves ω_0
  - double the spectral density
  - but also halves the magnitude of the coefficients
- In the limit, as $T_0 \to \infty$, $\omega_0 \to 0$ and $D_n \to 0$
  - the spectral components are spaced at zero (infinitesimal) intervals
  - magnitude of each component is zero
- An infinite amount of nothing!

Aperiodic Signal Representation

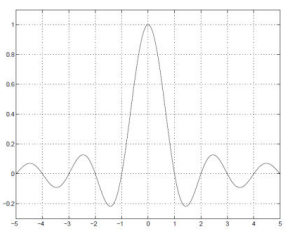
- To compensate, scale up by T_0:

$$\text{Scaled } D_n = T_0\frac{1}{n\pi}\sin\left(\frac{n\pi}{T_0}\right) = \frac{\sin(n\pi/T_0)}{n\pi/T_0}$$

- For large T_0 , can replace the densely packed discrete points n/T_0 with a continuous variable
  - say ω where $\omega = n2\pi/T_0$:

$$\text{Scaled } D_n \cong \frac{\sin(\omega/2)}{\omega/2} = \text{sinc}\,(\omega/2)$$

Sinc function



- Instead of a discrete set of spectral components, at the limit have a continuous frequency spectrum

The Fourier Transform

- If define a signal x_{T_0} formed by repeating an aperiodic signal $x(t)$ at intervals of T_0 seconds, and note that:

$$\lim_{T_0 \to \infty} x_{T_0}(t) = x(t)$$

- The Fourier series representing $x_{T_0}(t)$ will also represent $x(t)$ in the limit as $T_0 \to \infty$
- The exponential Fourier series for $x_{T_0}(t)$ is:

$$x_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \qquad D_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x_{T_0}(t) e^{-jn\omega_0 t} dt$$

The Fourier Transform

- As the period increases, the frequency components get closer, ultimately becoming a continuous variable:

$$\lim_{T_0 \to \infty} n\omega_0 = \omega \qquad (1)$$

- Multiplying D_n by T_0 and applying the limit in (1) gives:

$$\lim_{T_0 \to \infty} D_n T_0 = \lim_{T_0 \to \infty} T_0 \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x_{T_0}(t) e^{-jn\omega_0 t} dt = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (2)$$

- Limit is the ***Fourier transform*** of the time function $x(t)$, denoted by $X(\omega)$:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Inverse Fourier Transform

- To reconstitute $x(t)$ from $X(\omega)$ - need the *inverse transform*
- Re-write Fourier series representation for $x(t)$ as:

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} D_n T_0 e^{jn\omega_0 t} \frac{1}{T_0} \qquad (3)$$

- For $T_0 \to \infty$,
  - spacing between adjacent harmonics, $\Delta\omega = 2\pi/T_0$ approaches a differential separation $d\omega$, thus

$$\lim_{T_0\to\infty} \Delta\omega = d\omega \;\Rightarrow\; \lim_{T_0\to\infty} \frac{2\pi}{T_0} = d\omega \;\Rightarrow\; \lim_{T_0\to\infty} \frac{1}{T_0} = \frac{1}{2\pi} d\omega \qquad (4)$$

Inverse Fourier Transform

- Substituting the limits from (1), (3) and (4) into (2) and noting that the summation approaches an integral as $T_0 \to \infty$:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

- This is the ***inverse Fourier transform*** of the function $X(\omega)$
- $x(t)$ and $X(\omega)$ are a Fourier transform pair
  - where $X(\omega)$ is the frequency-domain specification of $x(t)$
- Note: the Fourier transform also represented by the operator $\mathcal{F}$:

$$X(\omega) = \mathcal{F}[x(t)] \quad \text{and} \quad x(t) = \mathcal{F}^{-1}[X(\omega)]$$

Alternative Notation

- If the frequency variable is in units Hz (denoted by f)
  - instead of rad/s (denoted by ω)

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$
$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

- Factor of $1/2\pi$ removed the inverse Fourier transform because $\omega = 2\pi f$, hence $d\omega = 2\pi df$

Frequency Spectrum

- Spectrum of a **periodic** signal is a **discrete** set of frequencies
  - possibly an infinite set (when there is a discontinuity)
- The Fourier transform of an **aperiodic** signal produces a **continuous** spectrum
- *Note:* Not all frequencies occur
  - i.e. $X(\omega)$ might be zero for some values of ω
- In particular, all frequencies outside of a certain range may be zero
  - e.g. $X(\omega) = 0$ for $|\omega|$ large
- Called ***band-limited signals*** - an important special class of signals
  - *(more details when sampling theory covered)*

Frequency Spectrum

- Can plot the spectrum $X(\omega)$ as a function of ω
- Since $X(\omega)$ is complex, have both amplitude and phase spectra:

$$X(\omega) = |X(\omega)|e^{j\angle X(\omega)}$$

- Note: for real $x(t)$, can show that:

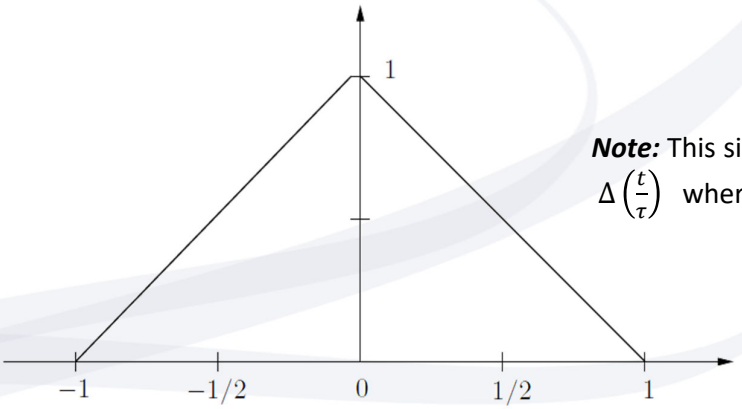
$$|X(-\omega)| = |X(\omega)|, \text{ and}$$
$$\angle X(-\omega) = -\angle X(\omega)$$

- Amplitude spectrum will always be an even function
- Phase spectrum will always be an odd function of ω
  - As seen in spectral plots from the exponential Fourier series of periodic signals

Example: Triangle Pulse

- Consider the Fourier transform of a triangle pulse defined by:

$$x(t) = \begin{cases} 1 - |t|, & |t| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



Note: This signal is commonly denoted as $\Delta\left(\frac{t}{\tau}\right)$ where $\tau = 2$ in this case

Example: Triangle Pulse

- For the Fourier transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-1}^{0}(1+t)e^{-j\omega t}dt + \int_{0}^{1}(1-t)e^{-j\omega t}dt$$

$$= \left(\frac{1+j\omega}{\omega^2} - \frac{e^{j\omega}}{\omega^2}\right) - \left(\frac{j\omega-1}{\omega^2} - \frac{e^{-j\omega}}{\omega^2}\right)$$

$$= \frac{e^{-j\omega}(e^{j\omega}-1)^2}{\omega^2} = \frac{e^{-j\omega}(e^{j\omega/2}(e^{j\omega/2}-e^{-j\omega/2}))^2}{\omega^2}$$

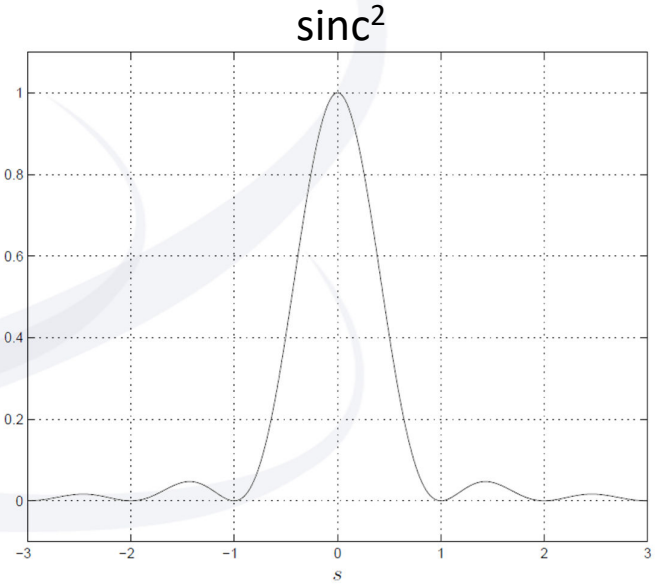
$$= \frac{e^{-j\omega}e^{j\omega}(2j)^2\sin^2(\omega/2)}{\omega^2}$$

$$= \left(\frac{\sin(\omega/2)}{\omega/2}\right)^2 = \mathrm{sinc}^2(\omega/2)$$

cf: Pair 19 in formula sheet

Example: Triangle Pulse

- Fourier transform of the triangle pulse is the square of the Fourier transform of a square pulse
  - No coincidence
- What do we get if we convolve a square pulse with itself?
- Relationship between convolution and the Fourier transform of the convolved signals?
  - More on this later!



Example: Exponential Decay

- The Fourier transform:

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t}dt = \int_{0}^{\infty} e^{-(a+j\omega)t}dt = \frac{-1}{a+j\omega}e^{-(a+j\omega)t}\Big|_{0}^{\infty}$$

- As $t \to \infty$, $e^{-(a+j\omega)t} \to 0$ for $a > 0$, therefore:

$$X(\omega) = \frac{1}{a+j\omega}, \qquad a > 0$$

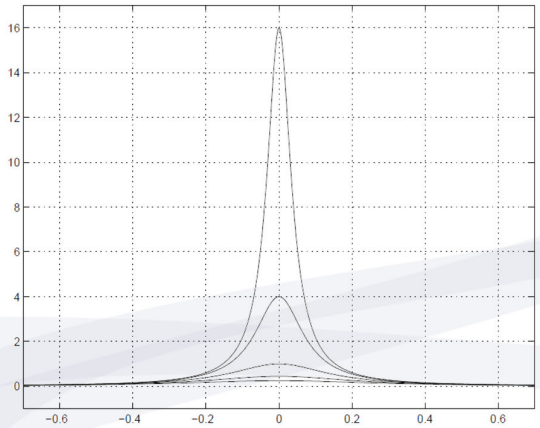
cf: Pair 1 in formula sheet

- The Fourier transform is complex
  - Unlike top hat and triangle pulses
    - Because the top hat and triangle pulses both had even symmetry
  - The exponential decay has no symmetry

Example: Exponential Decay

- The power spectrum of the exponential decay is:

$$|X(\omega)|^2 = \frac{1}{\left(\sqrt{a^2+\omega^2}\right)^2} = \frac{1}{a^2+\omega^2}$$



Power spectrum of $e^{-at}u(t)$ with a = 2, 1.5, 1, 0.5, 0.25

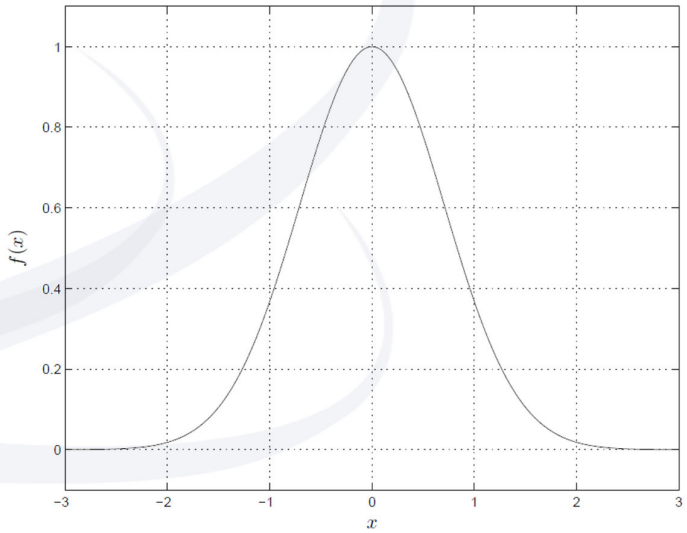
- Similar to a bell curve, but not Gaussian
- termed a Lorentz profile

Existence of the Fourier Transform

- Consider the case of $x(t) = e^{-at}u(t)$ when $a < 0$,
  - the Fourier integral does not converge
  - hence the Fourier transform does not exist for growing exponentials
- Not all signals are Fourier transformable
- The Fourier transform is a limiting case of the Fourier series
  - so the same basic conditions (suitably modified) apply
- Provided $x(t)$ satisfies the *Dirichlet conditions* , the Fourier transform is guaranteed to converge
- Any signal that can be generated in practice satisfies the Dirichlet conditions

Side Note: Gaussian Function

- An interesting Fourier transform example is that of the basic Gaussian function $f(x) = e^{-\pi x^2}$
- Can show that the Fourier transform of this is $\mathcal{F}[f(x)] = e^{-\pi x^2}$
- The Gaussian function is its own Fourier transform!
  - Interesting applications in image processing (and other areas)

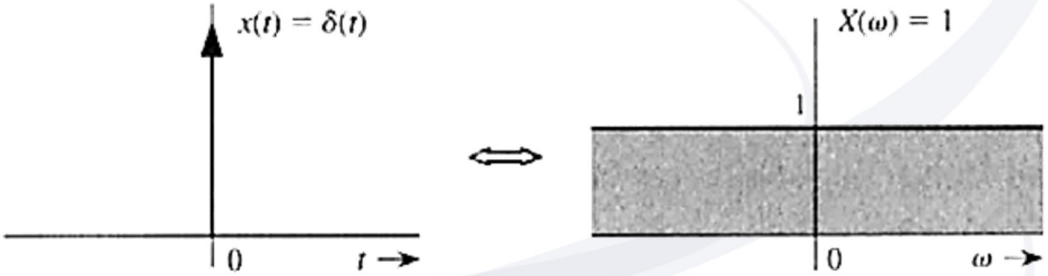


Example: Unit Impulse

- Using the sampling property of the impulse:

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

cf: Pair 6 in formula sheet



- The spectrum of the unit impulse contains frequency components at all frequencies!
  - the narrower the time-pulse, the wider the spectrum

Example: Unit Impulse

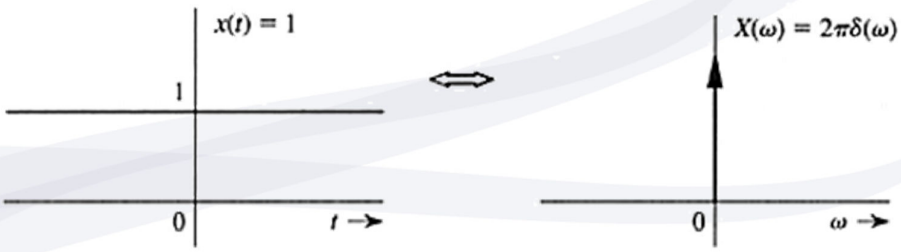
- The inverse transform of an impulse

$$\mathcal{F}^{-1}[\delta(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi}$$

$$\frac{1}{2\pi} \Leftrightarrow \delta(\omega) \quad \text{or} \quad 1 \Leftrightarrow 2\pi\delta(\omega)$$

cf: Pair 7 in formula sheet

- The spectrum of a constant (i.e. DC) signal $x(t) = 1$ is an impulse $2\pi\delta(\omega)$



DC signal has a single frequency $\omega = 0$

Example: Unit Impulse

- If frequency-shift the impulse

$$\mathcal{F}^{-1}[\delta(\omega-\omega_0)] = \frac{1}{2\pi}\int_{-\infty}^{\infty}\delta(\omega-\omega_0)e^{j\omega t}d\omega = \frac{1}{2\pi}e^{j\omega_0 t}$$

$$\frac{1}{2\pi}e^{j\omega_0 t} \Leftrightarrow \delta(\omega-\omega_0)$$

cf: Pair 8 in formula sheet

$$e^{j\omega_0 t} \Leftrightarrow 2\pi\delta(\omega-\omega_0) \quad \text{and} \quad e^{-j\omega_0 t} \Leftrightarrow 2\pi\delta(\omega+\omega_0)$$

- Spectrum of an everlasting exponential $e^{j\omega_0 t}$ is a single impulse at $\omega = \omega_0$
- What does this tell us about the spectrum of a sinusoidal signal?

Solution: Sinusoid

- For Fourier transform of the everlasting sinusoid $\cos\omega_0 t$:

$$\cos\omega_0 t = \tfrac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$$

From previous result:

$$\cos\omega_0 t \Leftrightarrow \pi[\delta(\omega+\omega_0) + \delta(\omega-\omega_0)]$$

cf: Pair 9 in formula sheet

- Spectrum of an everlasting sinusoid
  - only two frequency components at ω_0 and $-\omega_0$



Linearity of the Fourier Transform

- The Fourier transform operation is **linear**, i.e.:

  If $x_1(t) \Leftrightarrow X_1(\omega)$ and $x_2(t) \Leftrightarrow X_2(\omega)$

  then $a_1 x_1(t) + a_2 x_2(t) \Leftrightarrow a_1 X_1(\omega) + a_2 X_2(\omega)$

- ∴ Can use the Fourier transform to express a periodic signal as a sum of exponentials of the form $e^{jn\omega_0 t}$
- The Fourier series of a periodic signal $x(t)$ with period T_0 :

  $$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}, \quad \omega_0 = {}^{2\pi}/_{T_0}$$

- Taking the Fourier transform of both sides:

  $$X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} D_n \delta(\omega - n\omega_0)$$

Time–Frequency Duality

- Similarity between the Fourier transform operation and its inverse
- The two equations are **duals** of each other
- Only differences
  - change in variable from t to ω
  - the negative in the exponential term
  - scaling by 2π
- Gives rise to what is known as **time-frequency duality**
  - Example: the impulse Fourier transform
- This dual relationship applies to all aspects of these transforms

Time-Frequency Duality

- The duality property states that:

  If $x(t) \Leftrightarrow X(\omega)$, then, $X(t) \Leftrightarrow 2\pi x(-\omega)$

  or, if $x(t) \Leftrightarrow X(f)$, then, $X(t) \Leftrightarrow x(-f)$

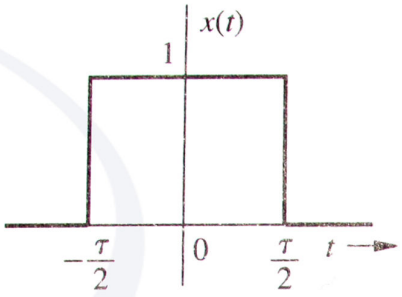
- This is apparent from the transforms:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(u)e^{jut}du$$

Hence,

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(u)e^{-jut}du$$

Time-Frequency Duality



- Generalising earlier result for the top hat function:

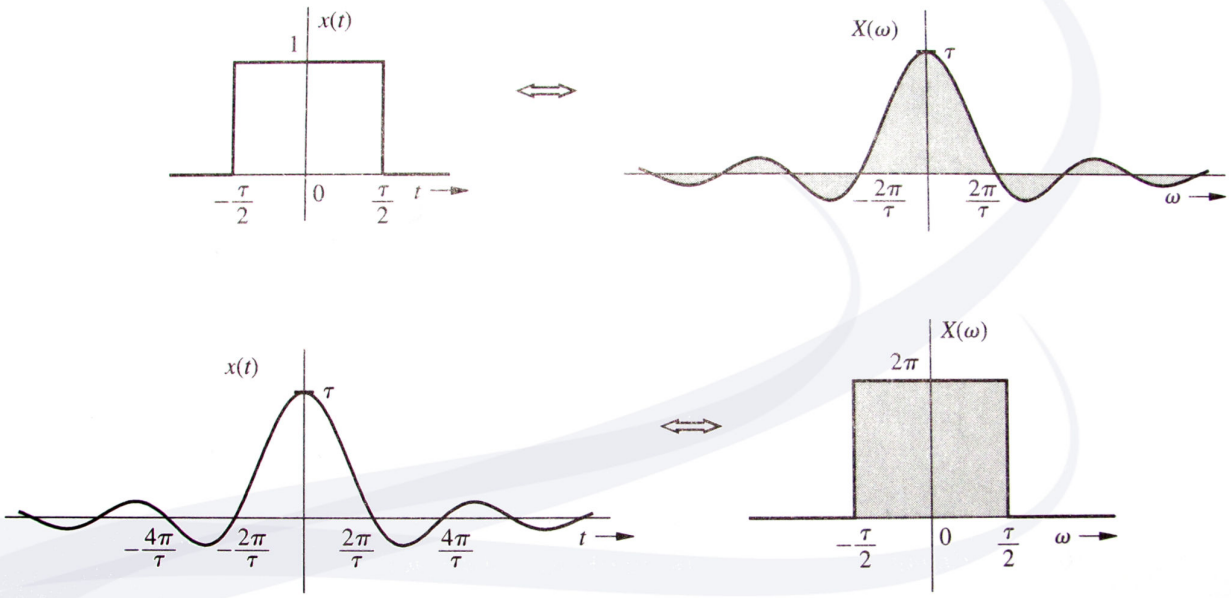
$$\underbrace{\text{rect}\left(\frac{t}{\tau}\right)}_{x(t)} \Leftrightarrow \underbrace{\tau\ \text{sinc}\left(\frac{\omega\tau}{2}\right)}_{X(\omega)}$$

- The duality property yields:

$$\underbrace{\tau\ \text{sinc}\left(\frac{t\tau}{2}\right)}_{X(t)} \Leftrightarrow \underbrace{2\pi\ \text{rect}\left(\frac{-\omega}{\tau}\right)}_{2\pi x(-t)} = 2\pi\ \text{rect}\left(\frac{\omega}{\tau}\right)$$

- Note, $\text{rect}\,(-x) = \text{rect}\,(x)$ due to the rect function being even

Time-Frequency Duality *



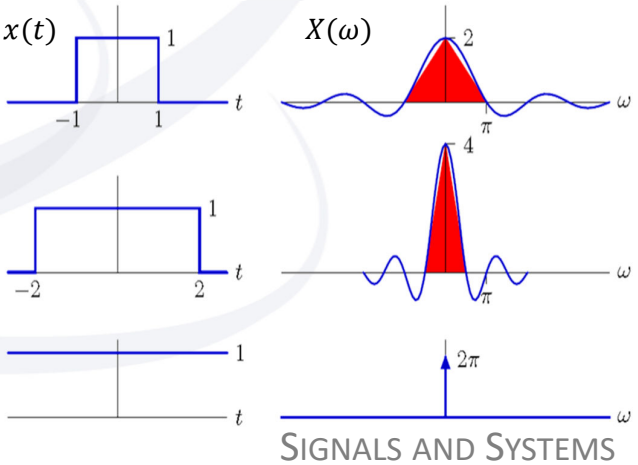
Scaling Property

- If $x(t) \Leftrightarrow X(\omega)$ then, for any real constant a:

$$x(at) \Leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

See textbook for proof

- Compression of a signal in time results in spectral expansion, and vice versa
- Compressing a signal in time → varies faster
  - ∴ higher frequencies are needed to synthesise it
  - frequency spectrum is expanded



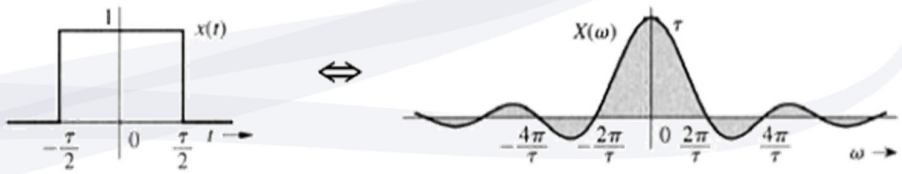
Scaling Property

- Scaling property: if $x(t)$ is wider, its spectrum is narrower, and vice versa
  - i.e. ***bandwidth of a signal is inversely proportional to the signal duration or width*** (in seconds)
- At the limit, leads to the ***Paley Wiener theorem:***
  - a signal that is time-limited cannot be band-limited
  - a signal that is band-limited cannot be time-limited
- Note:
  - ***Time-limited signal*** : signal that is non-zero only for a finite length of time
  - ***Band-limited signal*** : has a non-zero spectrum for a finite frequency interval

Solution Q2

- *Given the Fourier transform of a unit width rect function* $\mathcal{F}[rect\,(t)] = sinc\,(\omega/2)$*, what is the Fourier transform of a stretched rect of width* τ*?*
- A width τ rect function is $\text{rect}(t/\tau)$,
- Applying the scaling property: $x(at) \Leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$

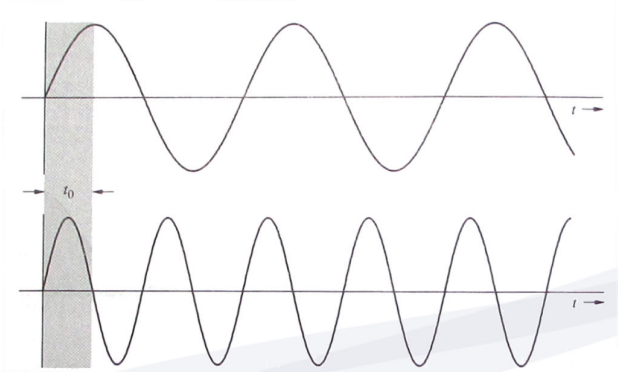
$$\mathcal{F}[\text{rect}\,(t/\tau)] = \tau\ \text{sinc}\,(\omega\tau/2)$$



Time-Shifting Property

ωt_0 phase shift

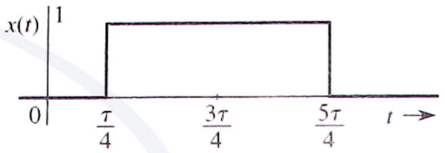
- If $x(t) \Leftrightarrow X(\omega)$ then $x(t - t_0) \Leftrightarrow X(\omega)e^{-j\omega t_0}$
- Consider two sinusoidal waves time shifted by the same t_0:



$$\cos(\omega(t - t_0) = \cos(\omega t - \omega t_0)$$

- **Amplitude is unchanged**
- But phase shift increases linearly with frequency

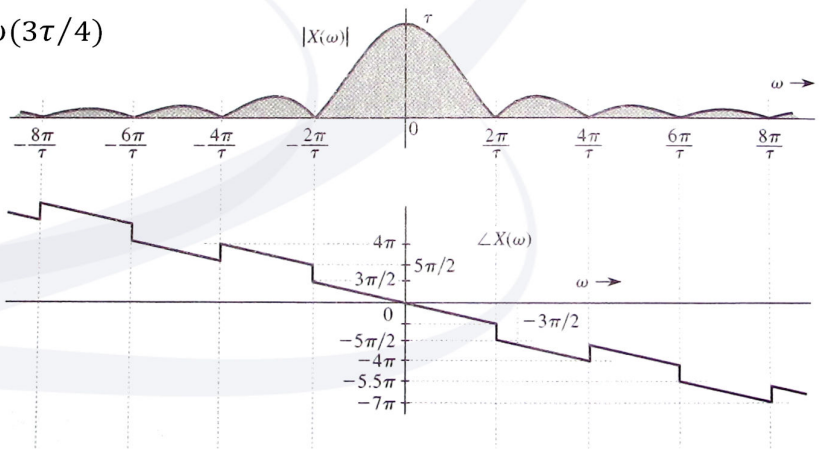
Time Shifting Example



- Find the Fourier transform of the shifted gate pulse shown above
- The pulse $x(t)$ is rect (t/τ) delayed by $3\tau/4$ s
- Hence, according to the time-shift property:

$$X(\omega) = \tau \text{ sinc } (\omega\tau/2)e^{-j\omega(3\tau/4)}$$

- Note: time-shift means the pulse no longer has even symmetry
  - So complex Fourier transform result



Frequency-Shifting Property

- If $x(t) \Leftrightarrow X(\omega)$ then $x(t)e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$

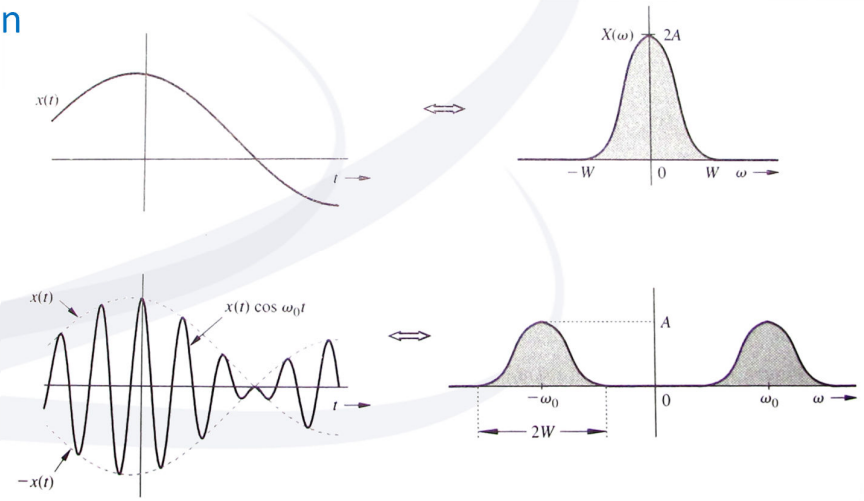
$$\mathcal{F}\left[x(t)e^{j\omega_0 t}\right] = \int_{-\infty}^{\infty} x(t)e^{j\omega_0 t}e^{-j\omega t}\,dt = \int_{-\infty}^{\infty} x(t)e^{j(\omega-\omega_0)t}\,dt = X(\omega - \omega_0)$$

- Multiplying a signal by a factor $e^{j\omega_0 t}$ shifts its spectrum by $\omega = \omega_0$
  - Note: duality between the time-shifting and frequency-shifting properties
- In practice: achieved by multiplying a signal by a sinusoid
  - As $e^{j\omega_0 t}$ is not a real function

$$x(t)\cos\omega_0 t = \frac{1}{2}\left[x(t)e^{j\omega_0 t} + x(t)e^{-j\omega_0 t}\right] \Leftrightarrow \frac{1}{2}\left[X(\omega - \omega_0) + X(\omega + \omega_0)\right]$$

Example: Amplitude Modulation

- Multiplication of a sinusoid $\cos\omega_0 t$ by $x(t)$
  - modulating the sinusoid amplitude by $x(t)$
- Known as amplitude modulation
  - sinusoid $\cos\omega_0 t$ is the carrier
  - $x(t)$ is the modulating signal
  - $x(t)\cos\omega_0 t$ is the modulated signal



Convolution Property

- **Convolution in time**: If $x(t) \Leftrightarrow X(\omega)$ and $h(t) \Leftrightarrow H(\omega)$, then:

$$x(t) * h(t) \Leftrightarrow X(\omega)H(\omega)$$

- Convolution in time is equivalent to multiplication in the frequency domain
- Note: Fourier transform of the impulse response is the system frequency response
- **Convolution in frequency**: If $x_1(t) = X_1(\omega)$ and $x_2(\omega) = X_2(t)$, then:

$$x_1(t)x_2(t) \Leftrightarrow \frac{1}{2\pi}X_1(\omega) * X_2(\omega)$$

- Time-frequency duality again

Time Differentiation and Integration

- **Time differentiation**: If $x(t) \Leftrightarrow X(\omega)$ then $\frac{dx}{dt} \Leftrightarrow j\omega X(\omega)$

Amplifies high frequencies

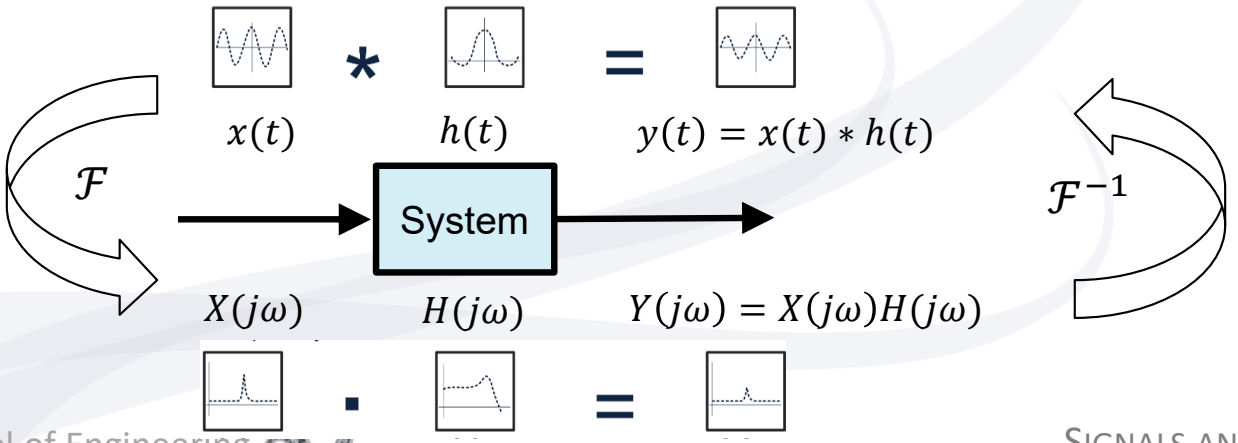
- **Time integration**: If $x(t) \Leftrightarrow X(\omega)$ then

$$\int_{-\infty}^{t} x(\tau)d\tau \Leftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$$

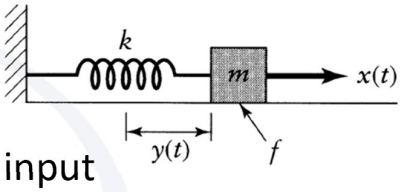
- Differentiation in time-domain = multiplication by $j\omega$ in frequency-domain
- Integration in time-domain = division by $j\omega$ in the frequency-domain
  - with the additional impulse term
    - represents the dc or average value resulting from the integration

Recap: LTI System Response

- If input signal frequency spectrum is $X(j\omega)$
- System frequency response, $H(j\omega)$, is known
- Then frequency spectrum of the output $Y(j\omega) = X(j\omega)\, H(j\omega)$



Example: Time Differentiation



- Mechanical system shown has an applied force $x(t)$ as its input
- Output is the position of the sliding mass $y(t)$
- Find: frequency response of the system, plot the magnitude response in dB
  - for *m* = 0.5 kg, *f* = 0.1 N.s/m and *k* = 50 N/m

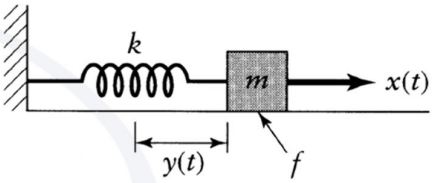
$$x(t) = m\frac{d^2}{dt}y(t) + f\frac{d}{dt}y(t) + ky(t)$$

Applying the time-differentiation property:

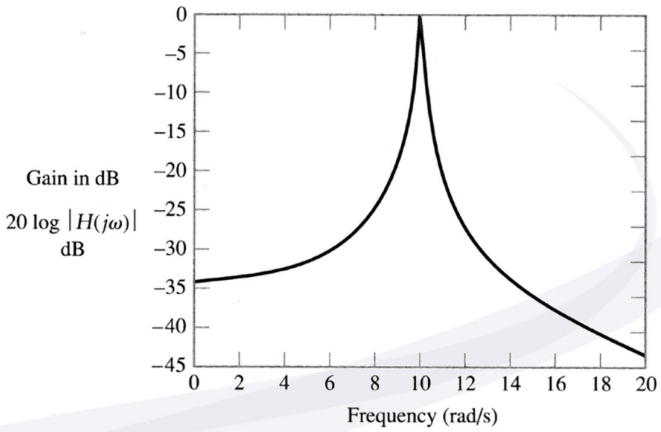
$$X(\omega) = m(j\omega)^2 Y(\omega) + fj\omega Y(\omega) + kY(\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{m(j\omega)^2 + fj\omega + k} = \frac{1}{-0.5\omega^2 + j0.1\omega + 50}$$

Example: Time Differentiation



- Magnitude plot of the frequency response, $|H(\omega)|$



- System acts as a very narrow band-pass filter
  - effectively only responds to frequency components near 10 rad/s:

Fourier Transform Operations Summary

Operation	$x(t)$	$X(\omega)$
Scalar multiplication	$kx(t)$	$kX(\omega)$
Addition	$x_1(t) + x_2(t)$	$X_1(\omega) + X_2(\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Scaling (a real)	$x(at)$	$\frac{1}{\lvert a\rvert} X\left(\frac{\omega}{a}\right)$
Time shifting	$x(t - t_0)$	$X(\omega)e^{-j\omega t_0}$
Frequency shifting (ω_0 real)	$x(t)e^{j\omega_0 t}$	$X(\omega - \omega_0)$

Also available in formula sheet

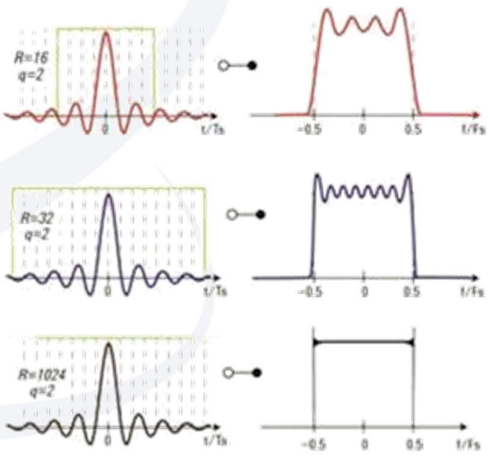
Fourier Transform Operations Summary

Operation	$x(t)$	$X(\omega)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(\omega)X_2(\omega)$
Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi}X_1(\omega) * X_2(\omega)$
Time differentiation	$\frac{d^n x}{dt^n}$	$(j\omega)^n X(\omega)$
Time integration	$\int_{-\infty}^{t} x(u)du$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$

- See also the table of Fourier transform pairs in the formula sheet

Fourier Transform Spectrum Properties

- **Property I:** Real and even signals produce real and even Fourier transforms
- **Property II:** Real and odd signals produce imaginary and odd Fourier transforms
- **Property III:** The amplitude and phase spectra of real signals are even and odd respectively
- **Property IV (Gibbs phenomenon):** The inverse Fourier transform of the truncated spectrum of a signal exhibits the Gibbs phenomenon at any signal discontinuities



System Response in Time vs Freq Domain

Impulse Response

$$\delta(t) \Rightarrow h(t)$$

$x(t)$ as sum of impulse components

$$x(t) = \int_{-\infty}^{\infty} x(t)\delta(t-\tau)d\tau$$

$y(t)$ as sum of responses to impulse components

$$y(t) = \int_{-\infty}^{\infty} x(t)h(t-\tau)d\tau$$

Response to everlasting exponential

$$e^{j\omega t} \Rightarrow H(\omega)e^{j\omega t}$$

$x(t)$ as sum of everlasting exponential components

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$$

$y(t)$ as sum of responses to exponential components

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)H(\omega)e^{j\omega t}d\omega$$

System Frequency Response *

- For a system with frequency response $H(\omega)$:

$$Y(\omega) = X(\omega)H(\omega) \qquad (y(t) = x(t) * h(t))$$

- In polar form:

$$|Y(\omega)|e^{j\angle Y(\omega)} = |X(\omega)||H(\omega)|e^{j[\angle X(\omega)+\angle H(\omega)]}$$

$$|Y(\omega)| = |X(\omega)||H(\omega)|, \quad \angle Y(\omega) = \angle X(\omega) + \angle H(\omega)$$

- Input signal's spectral component of frequency ω is modified in amplitude by a factor $|H(\omega)|$ and is shifted in phase by an angle $\angle Y(\omega)$
  - some components may be boosted in amplitude, others attenuated
  - phases may be shifted

Distortionless Transmission

- What is characteristic of system that allows signals to pass without distortion?
- Transmission is *distortionless* if the output is identical to the input within a multiplicative constant (relative delay is allowed):

$$y(t) = G_0 x(t - t_d) \quad \Rightarrow \quad Y(\omega) = G_0 X(\omega) e^{-j\omega t_d}$$

- But, $Y(\omega) = X(\omega)H(\omega)$, thus:

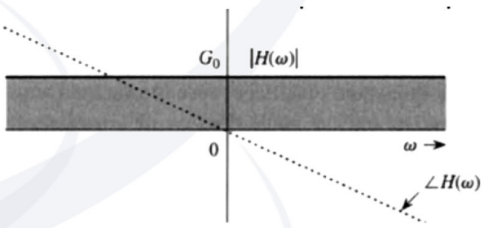
$$H(\omega) = G_0 e^{-j\omega t_d} \quad \Rightarrow \quad |H(\omega)| = G_0 \text{ and } \angle H(\omega) = -\omega t_d$$

- For distortionless transmission, the amplitude response $|H(\omega)|$ must be a constant AND phase response must be a linear function of ω with slope $-t_d$

Group Delay

- $\angle H(\omega) = -\omega t_d$: **every** spectral component is delayed by the same t_d seconds (irrespective of frequency)
- A distortionless transmission system needs a **flat** amplitude response and a **linear** phase response
- A measure of phase linearity is:

$$t_g(\omega) = -{}^{d}/_{d\omega} \angle H(\omega)$$

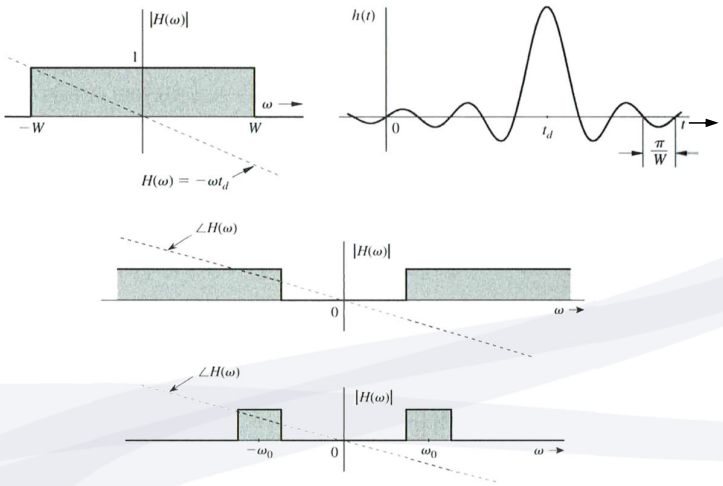


- $t_g(\omega)$ is known as ***group delay***, if it is a constant then all components are delayed by the same t_g seconds (assuming constant $\angle H(\omega)$)

Human ears are sensitive to amplitude distortion, but not phase distortion
Human eyes are sensitive to phase distortion, and less to amplitude distortion

Ideal vs Practical Filters

- Ideal filters allow distortionless transmission of a certain band of frequencies and completely suppress all others



Ideal low-pass filter

Components below $\omega = W$ rad/s passed, components above W suppressed

Ideal high-pass filter

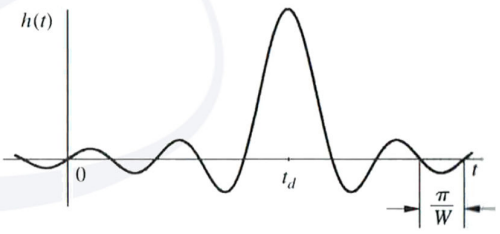
Ideal band-pass filter

Ideal vs Practical Filters

- The ideal low-pass filter:
  - delay an input signal bandlimited to W rad/s by t_d seconds
  - but otherwise pass it without distortion
- For this filter $H(\omega) = \text{rect}\left(\frac{\omega}{2W}\right) e^{-j\omega t_d}$
- Taking the inverse Fourier transform (using pair 18 and the time-shifting property):

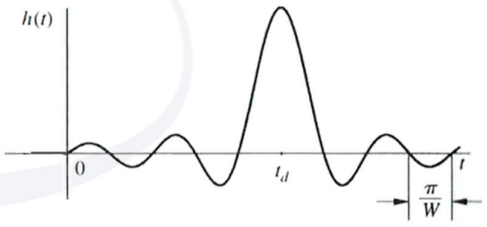
$$h(t) = \frac{W}{\pi} \text{sinc}\left[W(t - t_d)\right]$$

- Is $h(t)$ causal?
- No! The response begins before the impulse is applied
  - clearly not a physically realisable filter



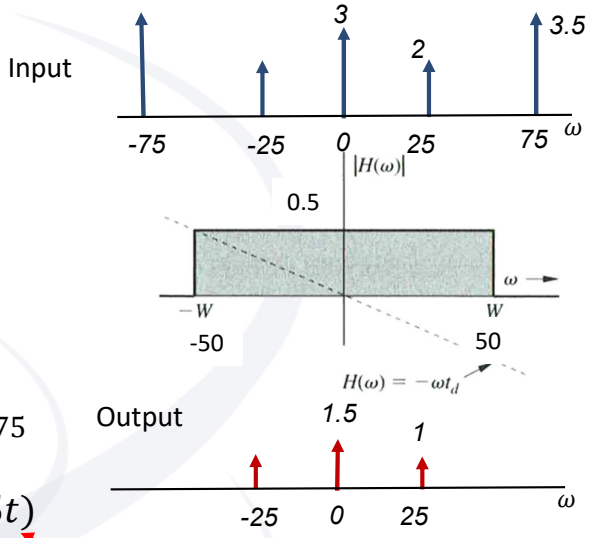
Ideal vs Practical Filters

- For a physically realisable system, $h(t)$ must be causal
- A practical approach to filter design: cut off the tail of $h(t)$ for $t < 0$,
  $\hat{h}(t) = h(t)u(t)$,
- If t_d is sufficiently large, $\hat{h}(t)$ will be a close approximation of $h(t)$
  - resulting filter $\hat{H}(\omega)$ a good approximation of an ideal filter
- The price of a close to ideal realisation:
  - a higher delay in the output
  - theoretically $t_d = \infty$ is required for an ideal filter
    - in practice a delay of 3 to 4 ($^{\pi}/_{W}$) will give a good approximation



Solution Q3

- Ideal filter has a frequency response given by
  $$H(\omega) = 0.5\ \text{rect}\left(\frac{\omega}{100}\right) e^{-j\omega(^{\pi}/_{10})}$$
- Passband magnitude = 0.5
- Bandwidth W = 50



$\omega = 0$ $\omega = 25$ $\omega = 75$

Input: $x(t) = 3 + 4\cos(25t) + 7\cos(75t)$

Filtered out

$|H(0)| = |H(25)| = 0.5$ $|H(75)| = 0$

$\angle H(0) = 0$ $\angle H(25) = -{}^{25\pi}/_{10} = -2.5\pi = -0.5\pi$

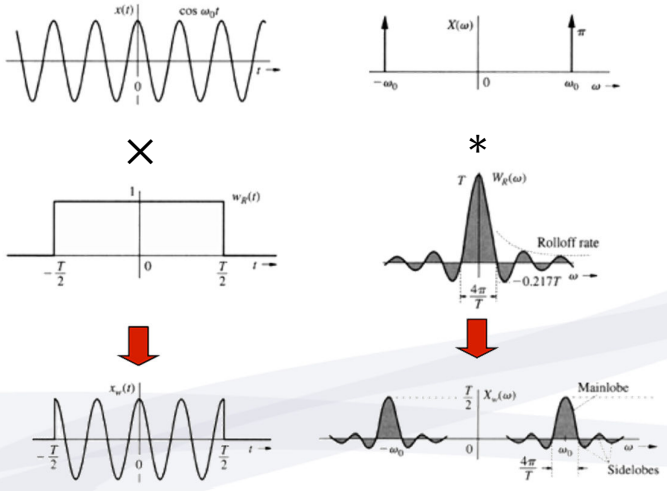
Output: $y(t) = 1.5 + 2\cos(25t - {}^{\pi}/_{2})$

Data Truncation: Windowing

- When representing signals for computational analysis
  - generally need to truncate the signal data
  - restricted to finite durations due to finite memory
- E.g. the impulse response $h(t)$ of an ideal low-pass filter
  - non-causal and approaches zero asymptotically as $|t| \to \infty$
- For practical design
  - must be truncated beyond a sufficiently large value of $|t|$
  - to make $h(t)$ causal and of finite duration
- Simplistic solution: ignore all data once values are sufficiently small
  - simple truncation – can cause some unexpected problems

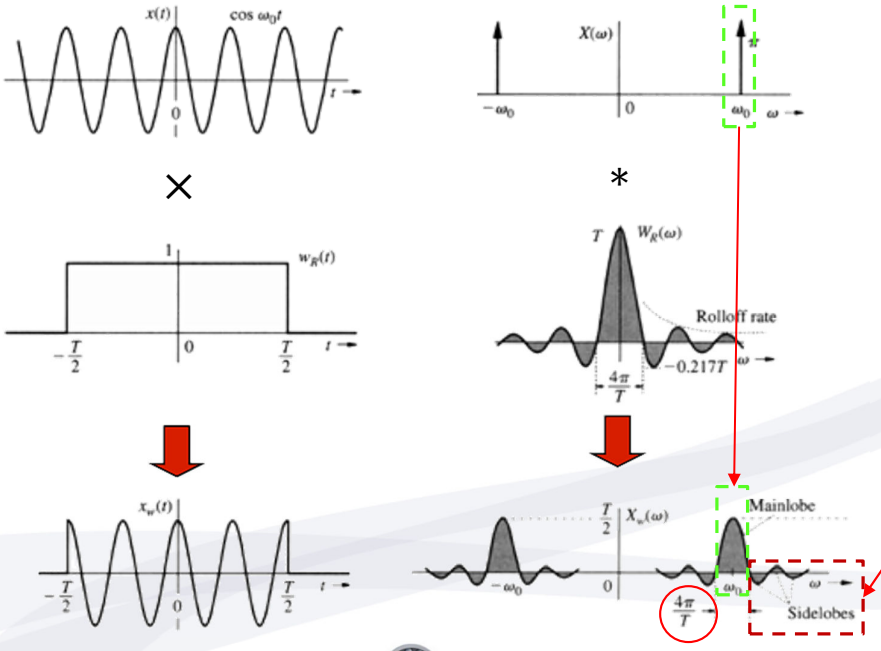
Data Truncation: Windowing

- Signal truncation: multiplying a signal of large (possibly infinite) width, by a window function of smaller (finite) width



- The simplest window function: rectangular window (straight truncation)
- What are the problems?
- **Spectral Spreading**: bandwidth increased by the bandwidth of $w(t)$
  - loss of spectral resolution
- **Leakage**: $W(\omega)$ not bandlimited
  - $X_w(\omega)$ spreads into the band where it should be zero

Data Truncation: Windowing



Spectral Spreading:

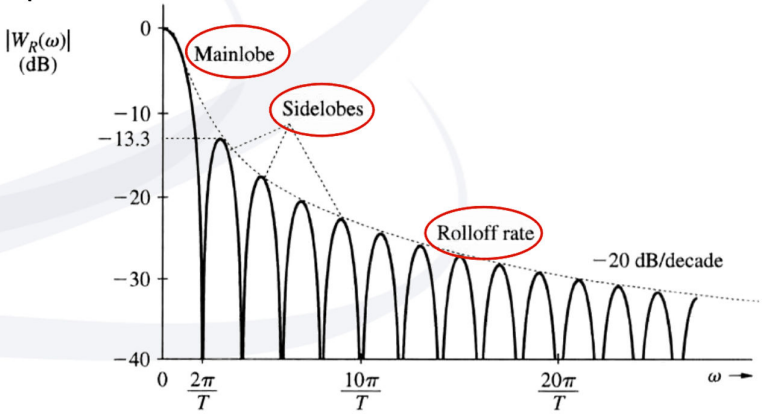
- Single frequency 'smeared' to range of frequencies $4\pi/T$ wide

Leakage:

- sidelobes – decaying
- Extend into band where it should be zero

Data Truncation: Windowing

- Spectral spreading is a problem
- If $x(t)$ has two spectral components of frequency less than $4\pi/T$ rad/s apart
  - Will be indistinguishable in the truncated signal
  - Ideally, spectral spreading as small as possible
- Also, for a rectangular window
  - the sidelobes only decay (*rolloff*) at a rate of $1/\omega$ (–20 dB/decade)
  - Prefer smaller sidelobes with a faster rolloff rate

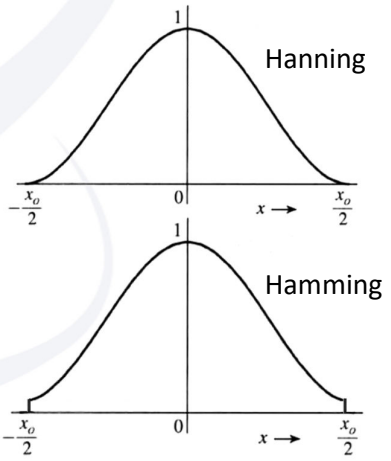


Potential Remedies

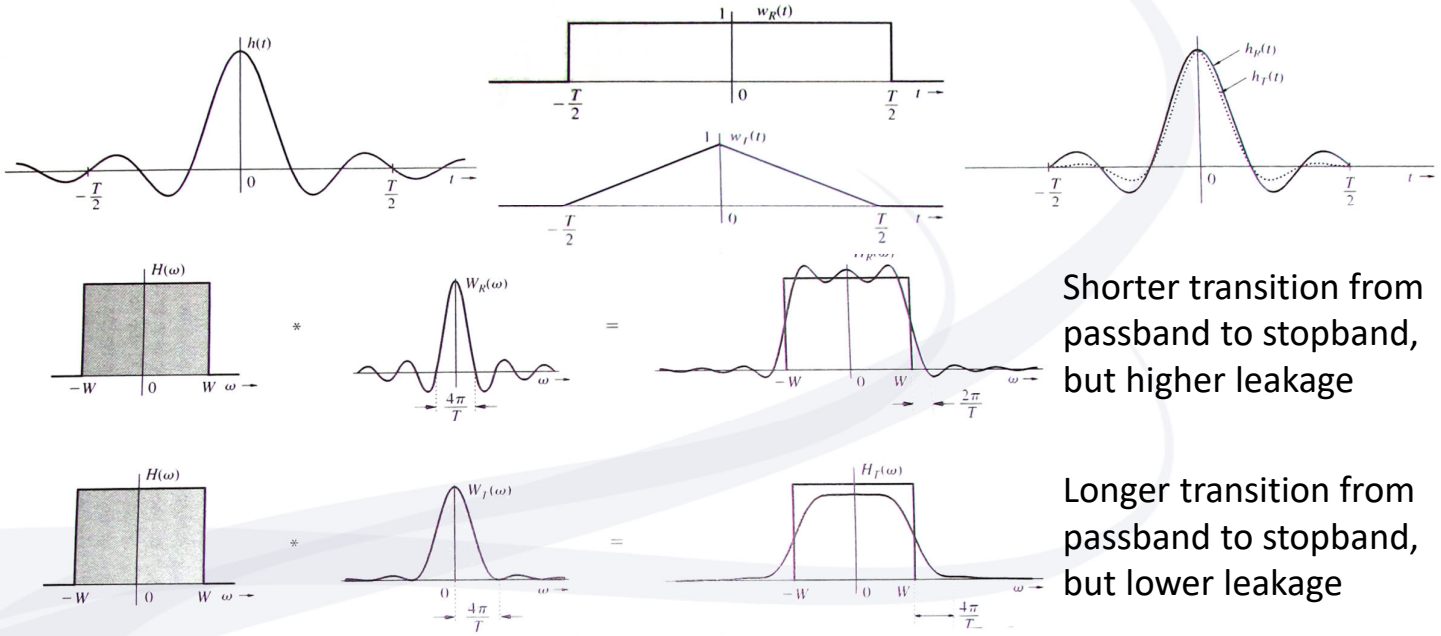
- For better results, need to minimise spectral spreading and leakage
- Spectral spread = bandwidth of $w(t)$
  - inversely proportional to the signal width
  - need to maximise the window width
- Recall: the slow $1/\omega$ sidelobe decay is due to the jump discontinuities
- The smoother a signal the faster the decay of its spectrum
  - a smoother (tapered) window will help with leakage
- Problem: Improving one deteriorates the other
  - a rectangular window of a given width has the smallest spectral spread but the slowest decaying sidelobes
  - a tapered signal has less leakage, but a wider mainlobe

Window Functions

- Need a compromise
- Different window functions used depending on the situation
  - Examples: the Bartlett (triangular), Hanning, Hamming, Blackman and Kaiser, which truncate the data gradually
- Different trade-offs with respect to mainlobe width, peak sidelobe magnitude, and leakage roll-off rate
- All are even functions of t
  - So the phase shift is either 0 or π
  - to minimise the distortion in the phase response



Example: Rectangular vs Triangular



Shorter transition from passband to stopband, but higher leakage

Longer transition from passband to stopband, but lower leakage

Parseval's Theorem *(again)*

- The energy of signal $x(t)$ can be derived in the time- or frequency-domains:

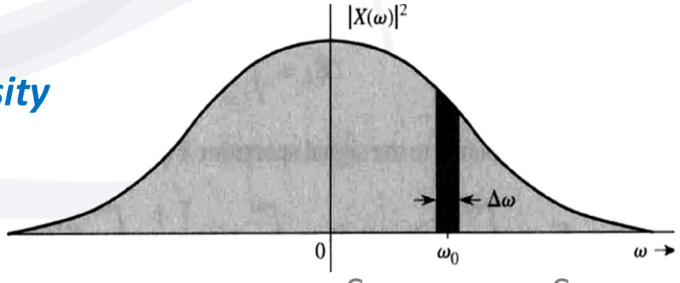
$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

See textbook for proof

- Thus the total signal energy is the area under $|X(\omega)|^2$ (divided by 2π)
- The energy over a small frequency band $\Delta\omega$ ($\Delta\omega \to 0$) is:

$$\Delta E_x = \frac{1}{2\pi} |X(\omega)|^2 \Delta\omega = |X(\omega)|^2 \Delta f$$

- Thus $|X(\omega)|^2$ is the ***energy spectral density***
  - per unit bandwidth in hertz



Energy Spectral Density of Real Signals

- For real signals $X(\omega)$ and $X(-\omega)$ are conjugates - $|X(\omega)|^2$ is an even function
- Thus, the signal energy can be shown as:

$$E_x = \frac{1}{\pi}\int_0^{\infty} |X(\omega)|^2 d\omega$$

- So, the energy contributed by a real signal by spectral components between ω_1 and ω_2 is:

$$E_x = \frac{1}{\pi}\int_{\omega_1}^{\omega_2} |X(\omega)|^2 d\omega$$

Example: Spectral Energy

- *Find the energy of signal $x(t) = e^{-at}u(t)$ and determine the frequency **W** (rad/s) so that the spectral energy of all components below **W** is 95% of the total signal energy E_x*
- Take FT of $x(t)$: $X(\omega) = {}^{1}/_{j\omega+a}$
- By Parseval's theorem:

$$E_x = \frac{1}{\pi}\int_0^{\infty} |X(\omega)|^2 d\omega = \frac{1}{\pi}\int_0^{\infty} \frac{1}{\omega^2 + a^2} d\omega$$

$$= \frac{1}{\pi a}\tan^{-1}\frac{\omega}{a}\Big|_0^{\infty} = \frac{1}{2a}$$

Example: Spectral Energy

- The band $\omega = 0$ to $\omega = W$ contains 95% of this signal energy, thus:

$$\frac{0.95}{2a} = \frac{1}{\pi}\int_0^W \frac{1}{\omega^2 + a^2}\,d\omega = \frac{1}{\pi a}\tan^{-1}\frac{\omega}{a}\bigg|_0^W = \frac{1}{\pi a}\tan^{-1}\frac{W}{a}$$

so

$$\frac{0.95\pi}{2} = \tan^{-1}\frac{W}{a} \;\Rightarrow\; W = 12.706\,a \text{ rad/s}$$

- Therefore: For this signal, 95% of the total energy is in a small frequency band from 0 to 12.7a rad/s or 2.02a Hz

Summary

- Have seen how the Fourier transform can be used to switch between the time- and frequency domains
  - and the duality between the two domains
- Seen how to represent and interpret the frequency spectra of signals and have shown that signal bandwidth is inversely proportional to signal width (duration)
- Explored the frequency response of systems and how aspects of ideality and finite data approximations impact system analysis and design